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FedHybrid-ViT: A Federated-Hybrid Framework for Generalizable Deepfake Detection on Heterogeneous Data

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Abstract
The rapid advancement of deep generative models has significantly increased the realism and accessibility of deepfake media, posing serious threats to digital integrity, public trust, and information security. Although recent deepfake detection methods report high accuracy on benchmark datasets, their performance often degrades under real-world conditions due to domain shift and data heterogeneity. Centralized models tend to overfit to dataset-specific artifacts and fail to generalize across diverse data sources, while federated learning (FL) approaches, although privacy-preserving, suffer from unstable convergence when trained on non-independent and identically distributed (non-IID) client data. To address these challenges, we propose FedHybrid-ViT, a privacy-aware deepfake detection framework that integrates a dual-stream Convolutional Neural Network (CNN) and Vision Transformer (ViT) within a federated learning setting. The CNN branch captures local high-frequency artifacts such as blending inconsistencies and compression traces, whereas the ViT branch models global semantic relationships across facial regions. The complementary features are fused to enable robust multi-scale representation learning. The framework is trained using a collaborative federated optimization protocol, allowing decentralized model updates without sharing raw data and thereby reducing the need for centralized raw data sharing. We evaluate the proposed approach on a heterogeneous multi-source dataset constructed from publicly available deepfake benchmarks under both independent and non-IID distribution settings. Experimental results demonstrate that FedHybrid-ViT consistently outperforms standalone CNN and ViT models as well as standard federated baselines. The proposed method achieves an AUC of 0.925 and maintains stable performance across highly heterogeneous conditions, indicating improved robustness under heterogeneous source distributions. These results highlight the effectiveness of combining hybrid architectures with federated learning for scalable and robust deepfake detection, offering a promising approach for distributed deepfake detection scenarios.

Keywords Deepfake detection · Federated learning (FL) · Domain generalization · Statistical heterogeneity · Non-IID data · Hybrid models · Vision transformer (ViT) · Privacy-preserving AI

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1 Introduction

The rapid evolution of deep generative models has democratized the creation of hyper-realistic synthetic media, widely known as deepfakes (Tolosana et al., 2020). While this has great potential in creative industries, the technology has also been weaponized to spread misinformation, manipulate public discourse, and violate individual privacy at a global scale. Therefore, the development of reliable automated detection systems is a task of paramount importance in digital media forensics (Verdoliva, 2020). As these manipulative techniques get more effective, the defense system has to develop from simple artifact detection to robust, privacy-aware systems operating in decentralized environments.

Existing studies have mainly targeted centralized deep learning paradigms, using CNNs and lately ViTs to detect manipulation traces. The state-of-the-art detectors showed unparalleled performance in controlled environments, consistently yielding AUC scores above 0.99 in popular benchmarks such as FaceForensics++ (Rossler et al., 2019). Because collecting a huge dataset of human faces in a central location brings about disturbing privacy concerns, the FL paradigm (Mcmahan et al., 2017) has been embraced, whereby collaborative model training is possible without exposing raw user data.

Despite these developments, we identify a fundamental gap in current deepfake detection methodologies, formalized as a "Dual-Failure": Centralized models suffer from brittle generalization; they overfit to specific source domains and fail catastrophically on "in-the-wild" data (domain shift). Standard federated approaches are prone to statistical heterogeneity. In practical scenarios, data across clients is non-independent and Identically Distributed (non-IID)—it varies by source characteristics, acquisition conditions, compression levels, and manipulation techniques. Under these conditions, conventional algorithms such as FedAvg (Mcmahan et al., 2017) and even personalized methods such as FedPer often result in model divergence and significant performance degradation (Zhao et al., 2018). To the best of our knowledge, existing studies have not jointly addressed global generalization and robustness to geographically heterogeneous non-IID distributions within a unified federated deepfake detection framework.

A central focus of this study is the effect of geographical heterogeneity as a practically important source of non-IID data in federated deepfake detection. The experimental setting comprises facial data from five geographically distinct regions, namely Africa, Asia, Europe, Korea, and the USA. Each regional collection represents a non-identical source distribution because of variations in source-level facial image characteristics, image acquisition conditions, compression characteristics, and manipulation patterns. These regional datasets are combined to construct a geographically heterogeneous multi-regional benchmark, after which the training subset is distributed across three logical clients to simulate a decentralized federated learning environment. In this study, the term non-IID primarily refers to the source-level statistical heterogeneity introduced by combining these non-identical regional distributions, rather than a strict one-region-per-client allocation. Because the client partitions are not explicitly balanced according to geographical region, their regional compositions may also vary. Although CNN, ViT, and conventional federated models may achieve strong performance on individual regional datasets, their performance can degrade on the combined non-IID benchmark because the substantially greater heterogeneity may encourage the learning of region- or dataset-specific features rather than manipulation-relevant forensic representations. Therefore, the primary novelty of this work is not merely

the integration of CNNs, ViTs, and Federated Learning, but the systematic investigation of deepfake detection under a geographically heterogeneous, source-level non-IID setting and the development of *FedHybrid-ViT*, which combines complementary local and global representations to improve robustness and generalization across diverse regional data distributions without requiring the centralized exchange of raw client data.

Our approach is motivated by the hypothesis that such a hybrid architecture can resolve these conflicting requirements by leveraging complementary feature representations. CNNs are intrinsically apt at detecting local high-frequency artifacts, such as blending boundaries, while ViTs excel at modelling long-range global dependencies and semantic inconsistencies. Integrating such streams within a federated framework allows us to learn a representation that is invariant to local data variance yet globally coherent. *FedHybrid-ViT* is designed to act like a resilient global learner and resist collapsing under the statistical chaos of the heterogeneous client data.

The key contributions of this work are summarized as follows:

- **Geographically Heterogeneous Evaluation:** We investigate federated deepfake detection using facial data from five geographically distinct regions, namely Africa, Asia, Europe, Korea, and the USA. These regional datasets are combined to construct a heterogeneous multi-regional benchmark, and the training subset is distributed across three logical clients for federated evaluation.
- **Hybrid Federated Framework:** We propose *FedHybrid-ViT*, a dual-stream federated framework that fuses the local forensic features extracted by ResNet-18 with the global semantic representations modelled by ViT-Base/16. This enables the model to capture both fine-grained manipulation artefacts and long-range semantic inconsistencies.
- **Robustness to Geographically Heterogeneous Non-IID Data:** We evaluate the proposed framework on a combined benchmark constructed from five non-identical regional source distributions and distributed across three logical federated clients. The results demonstrate that the complementary local–global representation of *FedHybrid-ViT* provides stronger robustness than CNN-only and ViT-only federated models when evaluated on the combined multi-regional non-IID dataset.
- **Comprehensive Comparative Evaluation:** With comprehensive evaluation against multiple centralized and federated baselines, we show that FedHybrid-ViT achieves an AUC of 0.925 among the evaluated configurations on the heterogeneous benchmark.
- **Robustness Across Multi-Regional Heterogeneous Data:** We provide empirical evidence that the proposed hybrid federated framework maintains stronger performance on the combined multi-regional heterogeneous distribution compared with its single-stream counterparts while reducing the need for centralized raw data aggregation.

The rest of the paper is organized as follows. Section II reviews related work on deepfake detection and Federated Learning. Section III describes the proposed architecture, feature-fusion mechanism, and federated training protocol. Section IV presents the experimental setup and results. Section V discusses the findings, and Section VI concludes the paper.

2 Related Work

Centralized deepfake detectors often deteriorate when trained on merged, multi-domain datasets. Their limitations stem not from insufficient data volume but from the inability to learn representations that generalize beyond dataset-specific artifacts. Three interrelated factors explain this behavior.

1. Overfitting to Dataset-Specific Artifacts: Deepfake detectors frequently rely on superficial visual cues unique to individual datasets—a phenomenon known as shortcut learning. Each dataset has its own distinctive "fingerprint," defined by generation pipelines, compression levels, and post-processing operations. For example, a detector trained solely on FaceForensics++ (FF++) may fixate on blending or boundary artifacts characteristic of that dataset, while one trained on Celeb-DF might internalize features related to its studio lighting and camera quality. When combined, such datasets form a "mixture of experts" whose artifact patterns conflict, making it difficult for a single model to discover a shared and generalizable notion of forgery.
2. Spurious Correlations and Source-Level Bias: Another obstacle to generalization arises from demographic and contextual biases. When regional datasets contain different acquisition conditions, generation characteristics, or source-specific patterns, models may inadvertently associate these non-causal characteristics with the real or fake label, leading to reduced robustness across heterogeneous data distributions.

To overcome these challenges, recent research has evolved along two largely separate directions. The first emphasizes architectural innovations—designing neural backbones that capture more universal forgery features. The second focuses on algorithmic improvements—optimizing the learning process to handle heterogeneous data in federated environments. This section reviews the most influential advancements in both domains, which form the baseline for the proposed FedHybridViT framework.

2.1 Architectural Solutions: The CNN vs. ViT Paradigm

Architectural strategies primarily address domain shift—the performance gap that occurs when a model trained on one dataset fails on another. The objective is to design backbones that extract features resilient to dataset-specific artifacts.

1. CNN-Based Models for Local Feature Detection Early breakthroughs in deepfake detection were powered by Convolutional Neural Networks (CNNs), such as XceptionNet (Chollet, 2017), EfficientNet (Tan & Le, 2019), and MesoNet (Afchar et al., 2018). CNNs are particularly good at detecting localized high-frequency inconsistencies such as compression anomalies, blending edges, and pixel-level distortions. Due to their strong inductive bias towards local patterns, they are well-suited for modeling the subtle mesoscopic cues introduced during face manipulation. Furthermore, several variants of CNNs operate in the frequency domain to amplify these subtle forgeries. However, while CNNs work well within familiar data, this locality bias limits the ability of the model to reason about global image semantics, thus making it vulnerable to unseen types of manipulations.

2. Vision Transformers for Global Context Modeling The rise of Vision Transformers (ViTs) (Dosovitskiy et al., 2021) brought with it a contrasting view. The models discard the convolutional locality bias and instead employ self-attention mechanisms that model global dependencies across the image. This allows them to capture broader semantic inconsistencies—such as unnatural head poses, mismatched lighting conditions, or incoherent facial dynamics—that CNNs usually fail to detect. Representatives from this category include the ViT-Base and Swin Transformer. However, since the ViTs do not possess inherent spatial inductive bias, they require large datasets and careful regularization to generalize well (Coccomini et al., 2022).
3. Hybrid CNN–ViT Architectures: The Current State-of-the-Art. Very recent works attempt to consolidate the merits of both architectures through hybrid CNNâ€“ViT models (Gu et al., 2021), where they combine local feature extraction along with global contextual reasoning. Some good examples include the GenConViT, Hybrid Transformer Network (HTN), and EfficientNet+ViT hybrids. Here, the CNN modules, such as those of EfficientNet-B4, act as feature extractors, which capture local texture and boundary artifacts, while the transformer modules analyze global spatial relationships among the extracted patches. Such hybrid backbones represent the current architectural state-of-the-art for achieving improved domain generalization, though they still remain confined to centralized training paradigms.

2.2 Algorithmic Solutions: Personalized Federated Learning (PFL)

Parallel to architectural developments, algorithmic approaches in Federated Learning (FL) aim to alleviate the challenge posed by the non-IID problem, where the client data distributions are significantly different. These techniques are developed with a focus on improving the aggregation strategy and inducing stable convergence even in the presence of data heterogeneity.

1. Standard Federated Averaging (FedAvg) The most basic baseline for FL is provided by FedAvg (Mcmahan et al., 2017), Here, each client first trains locally for several epochs, and then, the server aggregates the resulting weights by averaging them. Although effective under IID data, FedAvg's performance collapses under non-IID conditions because local updates diverge, leading to poor global optimization.
2. Regularization-Based Personalization: FedProx. To address the instability of FedAvg, FedProx (Li et al., 2020) introduces a proximal term into the local loss function. This regularization constrains local updates, preventing them from straying too far from the global model's parameters. The result is more consistent convergence in heterogeneous environments, albeit at the cost of slower adaptation for individual clients.

3 Methodology

This section details our proposed federated hybrid CNN-Vision Transformer (ViT) framework for deepfake detection. Our pipeline encompasses dataset construction, client-wise partitioning, hybrid architectural design, and a robust federated optimization strategy.

3.1 Framework Overview

We implement a four-stage workflow to address the challenges of generalizable deepfake detection. First, we convert a deepfake video corpus into a frame-level dataset of cropped face images with binary labels (real/fake). Second, we partition this data into multiple logical clients to simulate a decentralized federated setting. Third, we process each face through a dual-stream hybrid CNN-ViT model, extracting both local artifact features and global semantic representations. Finally, we train the model using a collaborative federated learning protocol that iteratively aggregates client-side updates.

3.2 Dataset Description and Distribution

To investigate the robustness of deepfake detection models under geographically heterogeneous facial distributions, we construct a multi-regional benchmark consisting of facial images collected from five geographically distinct regions: Africa, Asia, Europe, Korea, and the USA. The final dataset contains 51,222 facial images belonging to two classes: real and fake. Each regional subset is organized into separate real and manipulated image categories, where the directory-level class information is used as the ground-truth label. The dataset was constructed by combining publicly available deepfake samples from FaceForensics++ with additional internally collected facial samples prepared following the same binary real/fake labeling protocol. The regional categories were assigned according to the source characteristics of the collected samples. For publicly available samples, the original dataset annotations were preserved, whereas additional samples were manually organized into authentic and manipulated categories based on their generation process and source labels. The regional grouping was used only to analyze source-level variation and does not represent separate federated clients.

The proposed dataset contains substantially different regional distributions, which introduce source-level statistical heterogeneity. Specifically, the regional subsets contain variations in facial characteristics, demographic appearance, image generation patterns, and manipulation-related characteristics. Therefore, unlike conventional deepfake benchmarks that are primarily evaluated within a single dataset distribution, the proposed benchmark enables the analysis of model robustness under a geographically diverse setting.

The dataset distribution across the five regions is presented in Table 1. The USA subset contains the largest number of samples with 16,000 images, followed by Europe with 12,558 images, Korea with 8,926 images, Asia with 6,934 images, and Africa with 6,804 images. Although the regional sample sizes are not identical, each region contains a balanced or nearly balanced proportion of real and fake samples. Overall, the complete dataset contains 25,692 real images and 25,530 fake images, resulting in a balanced class distribution of 50.16% real samples and 49.84% fake samples.

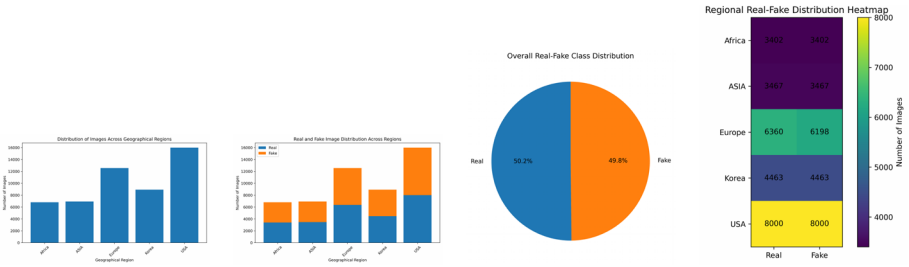
Table 1 Distribution of the proposed geographically heterogeneous deepfake dataset

Region	Real images	Fake images	Total images
Africa	3,402	3,402	6,804
Asia	3,467	3,467	6,934
Europe	6,360	6,198	12,558
Korea	4,463	4,463	8,926
USA	8,000	8,000	16,000
Total	25,692	25,530	51,222

The five regional subsets are first loaded independently and subsequently combined to construct the complete multi-regional benchmark. The combined dataset is then divided into training, validation, and testing subsets using class-stratified sampling to maintain the overall real–fake distribution. A fixed random state of 42 is used during dataset splitting. The resulting dataset partition consists of 64% training data, 16% validation data, and 20% testing data.

For federated evaluation, the training subset is distributed among three logical clients after combining all regional datasets. The client allocation is performed through index-based partitioning and is not explicitly assigned according to geographical regions. Therefore, the non-IID characteristic in this study refers primarily to the source-level heterogeneity introduced by combining five non-identical regional distributions. This setting enables the evaluation of federated deepfake detection models under a realistic heterogeneous facial data environment without requiring centralized collection of all raw data.

It is important to clarify that the federated clients in this study do not represent individual geographical regions. Instead, the regional datasets are first combined to construct a heterogeneous multi-regional benchmark, and the resulting training data are subsequently divided among three logical clients. Therefore, the federated setting represents a distributed learning scenario where clients contain heterogeneous samples originating from multiple regional sources. This design allows us to evaluate the robustness of federated optimization under source-level statistical variation rather than simulating independent geographical data owners. The distribution characteristics of the proposed geographically heterogeneous deepfake dataset are illustrated in Fig. 1. Figure 1a presents the sample distribution across the five geographical regions, demonstrating the diversity of the collected facial data. Figure 1b shows the real and fake sample distribution within each region, indicating a balanced class composition across different regional subsets. Figure 1c summarizes the overall class distribution, where real and manipulated samples remain nearly balanced. Finally, Fig. 1d provides a regional composition overview, highlighting the heterogeneous nature of the benchmark used for evaluating federated deepfake detection models.



(a) Regional image distribution (b) Real-fake distribution (c) Overall class balance (d) Regional composition heatmap

Fig. 1 Visualization of the proposed geographically heterogeneous deepfake dataset. **a** Image distribution across five geographical regions, **b** real and fake sample distribution across regions, **c** overall class balance, and **d** regional real-fake composition

3.3 Preprocessing and Feature Preparation

We process each raw frame through a face-detection backbone (e.g., MTCNN Zhang et al. 2016) to localize the facial region. We crop the detected face with a margin to preserve contextual features (such as the hairline and jaw) and resize it to 224×224 pixels as illustrated in Fig. 2.

For the CNN branch, we normalize images channel-wise using standard ImageNet mean and standard deviation. During training, we apply spatial and photometric augmentations—including random horizontal flips, rotations, and color jitter—to enhance robustness against pose and illumination variations.

Conversely, for the ViT branch, we process the resized image using the specific image processor associated with the ViT-Base/16 backbone. This ensures pixel-value scaling aligns with the transformer's pre-training configuration. We define a hybrid dataset object that returns a triplet $(x_i^{cnn}, x_i^{vit}, y_i)$ for each index i, ensuring synchronized data loading for both streams.

3.4 Hybrid CNN–Vision Transformer Architecture

Our proposed architecture employs a dual-stream network that processes face images through parallel convolutional and transformer pathways, as illustrated in Fig. 3. This design leverages complementary strengths: convolutions efficiently extract local, high-frequency artifacts, while transformers excel at modeling long-range dependencies and global facial semantics.

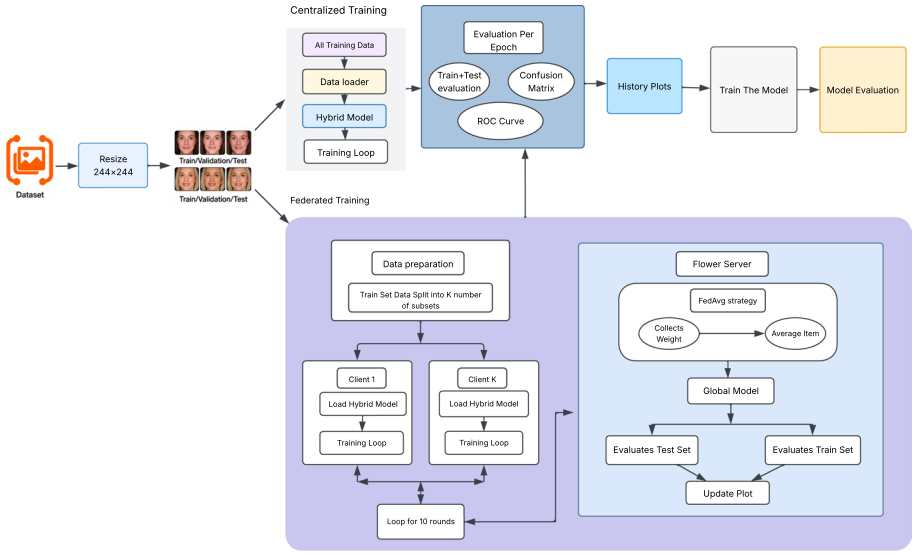


Fig. 2 Comprehensive workflow of the proposed deepfake detection framework. The diagram contrasts the centralized training approach with the federated training strategy, illustrating dataset preprocessing, hybrid model training at client nodes, and global model aggregation through the FedAvg strategy

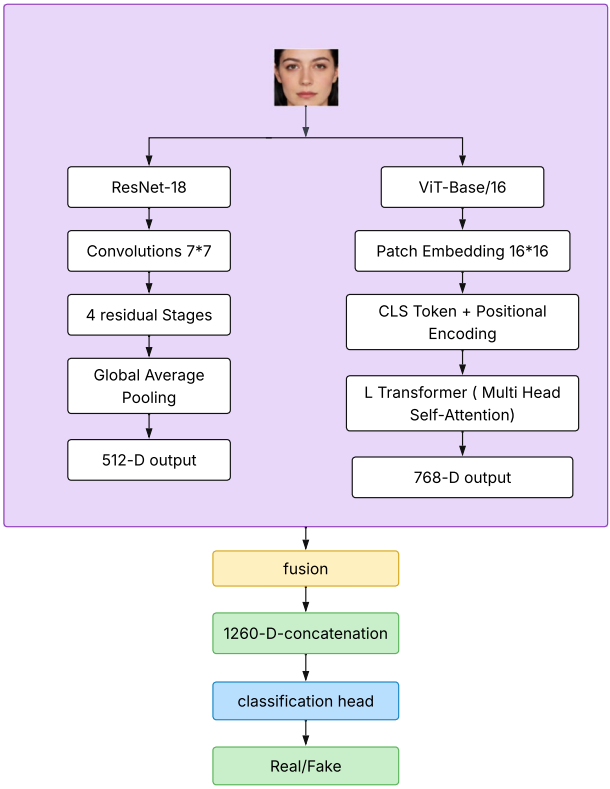


Fig. 3 Detailed architecture of the Hybrid CNN-ViT model. The input face is processed in parallel by ResNet-18 (capturing local features) and ViT-Base/16 (capturing global semantic features). The resulting descriptors are concatenated and passed through a classification head to determine authenticity

3.4.1 CNN Branch: Local Artifact Modeling

Let $x \in \mathbb{R}^{3\times224\times224}$ denote a preprocessed RGB face image. In the CNN branch, we employ a ResNet-18 backbone $f_{cnn}(\cdot;\theta_{cnn})$ pre-trained on ImageNet. The network comprises an initial 7×7 convolution followed by four residual stages. We pass the final feature map F_{cnn} through global average pooling to obtain a fixed-length descriptor:

$$h_{cnn} = g_{gap}(f_{cnn}(x)) \in \mathbb{R}^{d_{cnn}}, \quad d_{cnn} = 512 \tag{1}$$

We utilize this branch to capture localized inconsistencies typical of manipulation pipelines, such as blending seams and codec distortions, which residual CNNs detect effectively due to their inductive bias toward local patterns.

3.4.2 ViT Branch: Global Semantic Modeling

In parallel, we process the input x through a Vision Transformer branch $f_{vit}(\cdot;\theta_{vit})$. We partition the image into non-overlapping patches of size $P \times P$ (where $P = 16$), yielding N patches. We flatten and project each patch into a D-dimensional embedding:

$$z_i = W_p \text{vec}(x_i) + b_p, \quad i = 1, \ldots, N \tag{2}$$

We prepend a learnable class token z_{cls} and add positional encodings. The resulting sequence is processed by L transformer encoder blocks. We extract the transformed class token from the final layer as the global descriptor:

$$h_{vit} = \hat{z}_{cls} \in \mathbb{R}^{d_{vit}}, \quad d_{vit} = 768 \tag{3}$$

This pathway integrates information over the entire face, modeling correlations between distant regions (e.g., eye-mouth coherence) to identify higher-order semantic anomalies.

3.4.3 Feature Fusion

Since the CNN and ViT branches generate complementary feature representations, their extracted descriptors are directly concatenated to form a unified feature vector:

$$h_{fuse} = [h_{cnn}; h_{vit}] \in R^{1280} \tag{4}$$

The fused representation is then passed through a fully connected classification head to predict the probability of real and fake classes.

3.5 Federated Learning Strategy

We adopt the standard cross-device federated learning setting (Mcmahan et al., 2017). Let D_k denote the local dataset at client k. At the beginning of communication round t, the server selects a subset of clients $S^{(t)}$ and broadcasts the global weights $w^{(t)}$.

Each selected client initializes their local model with $w^{(t)}$ and performs E local epochs of stochastic gradient descent. The client then returns the updated parameters $w_k^{(t)}$ to the server. We aggregate these updates using a weighted average:

$$w^{(t+1)} = \sum_{k \in S^{(t)}} \left(\frac{|D_k|}{\sum_{j \in S^{(t)}} |D_j|} \right) w_k^{(t)} \tag{5}$$

This process repeats for T rounds.

3.6 Training Configuration and Implementation Details

The proposed framework was implemented in Python using PyTorch and Torchvision for model development, Hugging Face Transformers for the Vision Transformer backbone, Flower for federated learning simulation, Albumentations for image preprocessing and augmentation, and Scikit-learn for dataset splitting and performance evaluation. The experiments were executed in a Kaggle notebook environment running Python 3.11.13 on an NVIDIA Tesla T4 GPU with CUDA support.

All images were processed through two synchronized input pipelines. For the ResNet-18 branch, images were resized to 224×224 pixels and normalized using the ImageNet channel-wise mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225]. During training, random horizontal flipping with a probability of 0.5 and random brightness–con-

trast augmentation with a probability of 0.2 were applied. The validation and test images were resized and normalized without augmentation.

The regional datasets were loaded from five geographical folders and combined into a single multi-regional dataset. The combined dataset was divided into training, validation, and test subsets using class-stratified sampling with a fixed `random_state` of 42. First, 20% of the complete dataset was reserved for testing. Subsequently, 20% of the remaining 80% was used for validation. Consequently, the effective proportions of the complete dataset were 64% for training, 16% for validation, and 20% for testing. The training subset was shuffled during centralized training, whereas the validation and test subsets were evaluated without shuffling.

The hybrid model consists of an ImageNet-pretrained ResNet-18 and a pretrained ViT-Base/16. The original classification layer of ResNet-18 was removed to obtain a 512-dimensional CNN feature vector. For the ViT branch, the 768-dimensional representation of the final `[CLS]` token was extracted. Both pretrained branches remained trainable throughout the experiments. The two feature vectors were directly concatenated to form a 1280-dimensional representation, which was passed to a fully connected layer with two output units corresponding to the real and fake classes. No additional batch-normalization, feature-projection, or intermediate fusion layer was implemented. Model training used the cross-entropy loss function.

For centralized training, the hybrid CNN–ViT model was trained for five epochs using the Adam optimizer with a learning rate of 1×10^{-5} and a batch size of 16. The default Adam parameters were used, including $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$. No explicit weight decay, learning-rate scheduler, or early-stopping strategy was employed. Training and validation performance were evaluated after every epoch, and the final model was subsequently evaluated on the held-out test set.

For federated evaluation, the training subset was divided into three approximately equal partitions using index-based splitting, with each partition assigned to one logical client. This allocation was performed after combining the regional datasets and was not explicitly stratified according to geographical region. Accordingly, the experimental setting captures the source-level non-IID heterogeneity of the combined multi-regional benchmark rather than enforcing a strict one-region-per-client allocation.

The federated learning simulation was implemented using the standard Federated Averaging (FedAvg) strategy. Training was performed for five communication rounds, and every client completed three local epochs during each round. All three clients participated in every communication round, corresponding to a client participation fraction of 1.0. Each client used the Adam optimizer with a learning rate of 1×10^{-5}, a batch size of 16, and the cross-entropy loss function. The local Adam optimizer was initialized at the beginning of each client-training operation. After local training, client model parameters were returned to the server and aggregated according to the number of training samples available at each client. No client-side federated evaluation was performed; instead, the aggregated global model was evaluated on the common held-out test set after each communication round. In the simulation environment, each client was allocated one-third of the available GPU resources.

Performance was measured using classification loss, accuracy, macro-averaged precision, macro-averaged recall, macro-averaged F1-score, receiver operating characteristic area under the curve (ROC-AUC), and confusion matrices. Accuracy was obtained from

the predicted class labels, whereas ROC-AUC was calculated using the softmax probability assigned to the fake class.

3.7 Evaluation Metrics

Model performance is evaluated on the held-out test set using standard binary classification metrics. Let TP, TN, FP, and FN denote the number of true positives, true negatives, false positives, and false negatives, respectively. The metrics are defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{6}$$

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \tag{7}$$

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{8}$$

In addition, the area under the receiver operating characteristic curve (AUC) is computed from the predicted softmax scores to capture the trade-off between true-positive and false-positive rates over all thresholds. Confusion matrices are reported to analyze error patterns between real and fake classes. For federated experiments, metrics are reported both globally, by aggregating predictions across all clients' test data, and per client, to quantify the impact of data heterogeneity on local performance.

3.8 Baselines and Comparison Setup

To isolate the contributions of the hybrid architecture and the federated optimization, the following baselines are implemented:

- **Centralized CNN:** ResNet-18 trained end-to-end on the centralized training set.
- **Centralized ViT:** ViT-Base/16 fine-tuned end-to-end on the centralized training set.
- **Centralized hybrid CNN–ViT:** the proposed dual-stream architecture trained with standard data parallelism, without federated partitioning.
- **Federated CNN:** ResNet-18 trained under the same federated protocol (same K, E, T, and client sampling) as the hybrid model.
- **Federated ViT:** ViT-Base/16 trained under the same federated protocol.
- **Federated hybrid CNN–ViT (proposed):** the full system described above.

All models use the same data splits, optimizer, batch size, and stopping criteria. For federated baselines, the same client-partitioning scheme, number of communication rounds, and local epochs are used to ensure that performance differences are attributable to architectural differences rather than training budgets.

3.9 Implementation and Reproducibility

The system is implemented in PyTorch, using official model definitions and pre-trained weights for ResNet-18 and ViT-Base/16. The federated learning setup is realized via a serverâ€“client abstraction in which each client is modeled as a process with its own data loader and optimizer, while the server maintains the global parameters and performs aggregation at the end of each communication round.

Experiments were conducted in a Kaggle notebook environment equipped with an NVIDIA Tesla T4 GPU with CUDA support. All random seeds (for data shuffling, weight initialization, and client sampling) are fixed to ensure deterministic runs where possible. Hyperparameters such as learning rates, batch sizes, local epochs E, number of rounds T, and client-partitioning regimes are recorded and kept consistent across repeated experiments.

To facilitate reproducibility, the codebase can be released with configuration files specifying dataset paths, model architectures, federated-training parameters, and evaluation scripts, enabling independent researchers to reproduce the centralized and federated results reported in the paper.

4 Results

In this section, we present a comprehensive evaluation of the *FedHybrid-ViT* framework. In this section, we assess the model's performance across diverse regional source distributions and evaluate its robustness on a combined heterogeneous dataset. Table 2 summarizes the accuracy of our proposed method against standalone CNN and ViT baselines under both centralized and federated training paradigms.

4.1 Performance on Regional (Homogeneous) Datasets

For the regional evaluation, each regional subset was evaluated independently using the corresponding test samples. These results represent the performance of models when evaluated under relatively homogeneous regional distributions. In contrast, the Combined Region

Table 2 Performance comparison including combined dataset accuracy

Model	Training type	Africa region	Asia region	Europe region	Korea region	USA region	Combined region
CNN	Centralized (Baseline)	94.3%	95.6%	96.9%	92.8%	97.9%	91.3%
CNN	Federated Learning	94.4%	96.7%	97.2%	93.6%	97.4%	93.1%
ViT	Centralized (Baseline)	96.6%	96.5%	98.1%	87.7%	97.7%	92.6%
ViT	Federated Learning	96.3%	97.6%	97.6%	90.1%	98.2%	92.0%
Hybrid (CNN+ViT)	Centralized (Baseline)	95.8%	97.4%	97.3%	92.9%	98.5%	95.1%
Hybrid (CNN+ViT)	Federated Learning (Proposed)	96.2%	97.1%	97.7%	93.2%	98.4%	94.6%

The proposed FedHybrid-ViT maintains high accuracy even when regions are combined, unlike centralized baselines

Remarks: Individual regional datasets achieve high performance for all models, while performance decreases on the combined heterogeneous benchmark. The proposed FedHybrid-ViT improves robustness under diverse data distributions and reduces the impact of domain shifts

evaluation uses the complete heterogeneous test set containing samples from all five regional sources and is used to analyze generalization under source-level distribution variation. We first analyze the performance on individual regional datasets (Africa, Asia, Europe, Korea, USA), which represent siloed homogeneous environments. As shown in columns 3 through 7 of Table 2, we observe that all models achieve relatively high accuracy when trained and tested within a specific region.

- The **Centralized CNN** and **ViT** baselines achieve accuracies ranging from 92.8% to 98.1%. This confirms that standard deep learning architectures can effectively learn deepfake artifacts when the data distribution is IID (Independent and Identically Distributed) and demographically consistent.
- Our **FedHybrid** approach remains highly competitive in these silos, achieving **98.4%** in the USA region and **97.7%** in the Europe region, demonstrating that our hybrid architecture does not sacrifice local specificity.

4.2 Generalization on the Combined (Heterogeneous) Dataset

The critical evaluation lies in the Combined Region column, which represents a geographically heterogeneous, source-level non-IID scenario constructed from five non-identical regional data distributions. Here, we observe a distinct performance degradation in the baseline models.

- The **Centralized CNN** drops to **91.3%**, and the **Federated ViT** struggles at **92.0%**. This supports our hypothesis that single-stream models struggle to generalize when exposed to diverse manipulation types and demographic variations simultaneously.
- In contrast, our proposed **FedHybrid-ViT** demonstrates improved robustness, achieving an accuracy of **94.6%** in the federated setting. Notably, it outperforms the Federated CNN (93.1%) and Federated ViT (92.0%) by significantly surpassing.
- Furthermore, our federated model performs comparably to the **Centralized Hybrid** baseline (95.1%). This minimal performance gap ($< 0.5\%$) indicates that the proposed hybrid federated model remains competitive with its centralized counterpart when trained and evaluated on geographically heterogeneous, source-level non-IID data.

4.3 Ablation Study

Although this analysis is referred to as an ablation study, it primarily evaluates the contribution of different model configurations rather than performing a complete component-wise ablation of every architectural element. In particular, the independent contribution of the feature fusion operation is not separately investigated. The objective of this analysis is to compare the effectiveness of different backbone combinations and training paradigms under the same heterogeneous federated setting.

To analyze the contribution of different components of the proposed framework, an ablation study is conducted by comparing CNN-only, ViT-only, CNN-ViT, and their federated variants. The objective is to evaluate the effectiveness of combining local convolutional features with global transformer-based representations under the heterogeneous federated setting.

The CNN-only and ViT-only models represent individual feature learning approaches, where CNN focuses on local manipulation artifacts and ViT captures global semantic dependencies. The hybrid CNN-ViT model improves the detection capability by combining these complementary representations. Furthermore, the federated experiments demonstrate the effect of decentralized training on these architectures.

As shown in Table 2, the proposed FedHybrid-ViT achieves the best performance among the federated configurations, demonstrating that the integration of CNN and ViT representations provides improved robustness compared with single-stream federated models. The results confirm that the local–global feature fusion is beneficial for deepfake detection under geographically heterogeneous non-IID data distributions.

4.4 Analysis of Precision, Recall, F1-Score, and AUC

Beyond accuracy, precision, recall, F1-score, and ROC-AUC provide additional insights into the behavior of the proposed model. The centralized hybrid model achieves a precision of 95.13%, recall of 95.10%, and F1-score of 95.10%, whereas FedHybrid-ViT obtains 94.59%, 94.55%, and 94.55%, respectively. The performance gap between centralized and federated training remains below 0.6 percentage points across all metrics, demonstrating that the proposed federated framework preserves comparable detection capability while operating in a distributed environment.

The precision results indicate that FedHybrid-ViT produces reliable fake-sample predictions with limited false alarms. Meanwhile, the recall performance confirms that the model can identify the majority of manipulated samples despite the heterogeneous client distribution. The comparable F1-score demonstrates a balanced trade-off between false positive and false negative errors. Furthermore, the ROC-AUC value of 0.925 indicates strong threshold-independent discrimination capability, confirming the effectiveness of the hybrid CNN-ViT representation for distinguishing real and fake facial images under geographically heterogeneous conditions.

The detailed comparison of classification metrics is provided in Table 3. Although centralized training achieves marginally higher performance, FedHybrid-ViT exhibits only a negligible degradation across all metrics. Specifically, the performance difference remains below 0.6 percentage points for accuracy, precision, recall, and F1-score. This demonstrates that the proposed federated framework can retain competitive detection capability compared with centralized training while operating under a decentralized learning setting.

4.5 Visual Analysis

To further analyze the specific error patterns, we visualize the confusion matrices and training history plots for all six experimental configurations on the combined dataset in Fig. 4.

Comparing the Centralized baselines (Fig. 4) with the Federated models (Fig. 4), we observe the impact of decentralized training. The **Federated ViT** exhibits a notable increase in false negatives compared to its centralized counterpart, confirming that pure Transformers struggle to converge without centralized data access. Conversely, our proposed **FedHybrid-**

Table 3 Performance comparison of centralized and federated hybrid models

Model	Acc.	Prec.	Rec.	F1	AUC
Centralized Hybrid CNN-ViT	95.10	95.13	95.10	95.10	0.951
FedHybrid-ViT	94.55	94.59	94.55	94.55	0.925

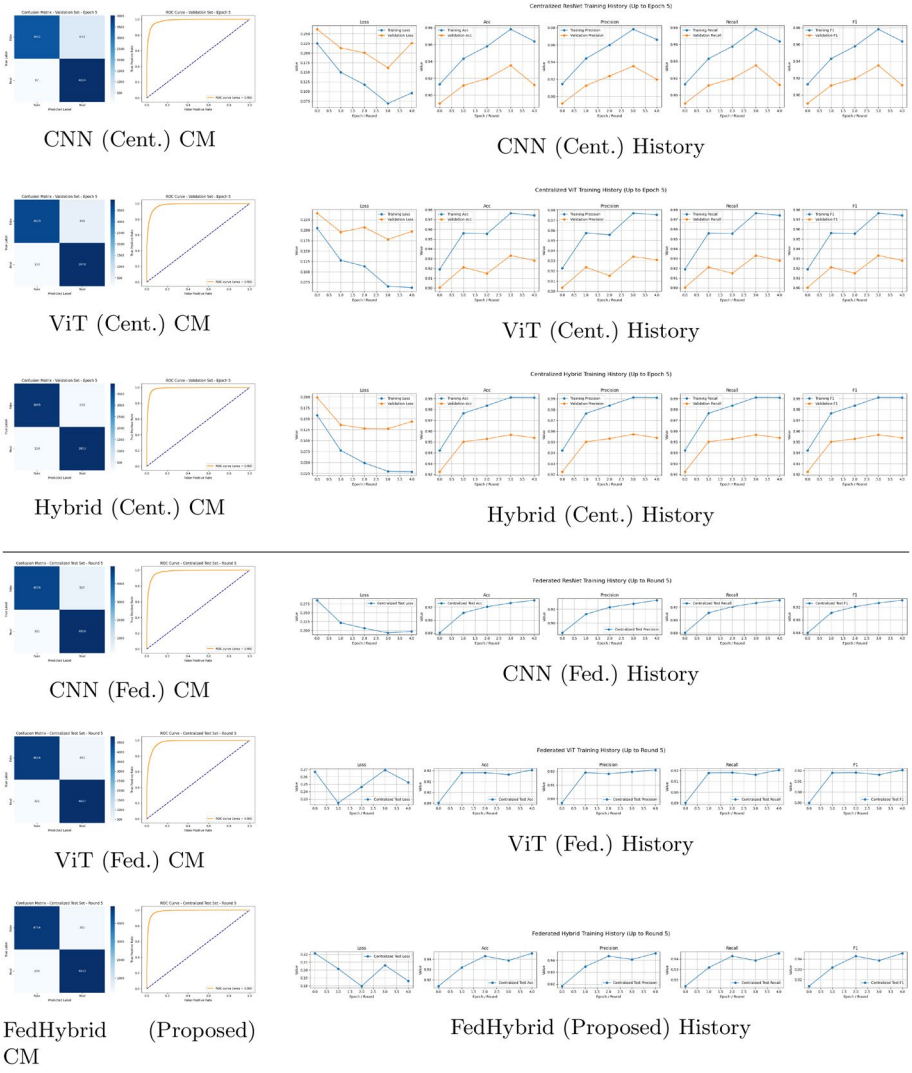


Fig. 4 Comprehensive visual analysis. The top three rows represent Centralized baselines, while the bottom three rows represent Federated settings. Left column: Confusion Matrices (CM). Right column: Training History. Note how the FedHybrid-ViT (bottom row) stabilizes the training curve and improves confusion matrix diagonal dominance compared to the standard Federated ViT

ViT (Bottom row of Fig. 4) maintains a strong diagonal structure almost identical to the **Centralized Hybrid**. This visual evidence confirms that our hybrid architecture effectively stabilizes the federated learning process, preventing the "collapse" seen in standard models.

5 Discussion

Our experimental results validate the efficacy of the *FedHybrid-ViT* framework in addressing the "Dual-Failure" of generalization and privacy. Here, we interpret these findings through the lens of our two primary contributions: the mitigation of data bias and the architectural advantage of the hybrid design.

5.1 Mitigating Data Bias and the Silo Effect

A major novelty of our research is the rigorous evaluation on a combined regional dataset. Most existing studies report high performance on single-source benchmarks. However, our results reveal that models performing exceptionally well in isolation (e.g., ViT Centralized at 98.1% in Europe) suffer from domain shift when applied to a global, combined dataset (dropping to 92.6%).

By training on combined regional data, we expose the models to a wider variance of facial features and environmental conditions. Our results show that while single-stream models struggle to adapt to this variance, the *FedHybrid-ViT* successfully synthesizes diverse features. This suggests that our framework prevents the model from overfitting to region-specific "shortcuts" (such as specific lighting or skin textures), resulting in a more unbiased and globally generalizable detector.

5.2 The Hybrid Advantage: Architecture Matters

The performance gap between our hybrid approach and the standalone baselines underscores the necessity of the proposed dual-stream architecture.

- **CNN Limitations:** We observe that the CNN baselines, while stable, plateau in performance on the combined dataset (91.3% centralized). This suggests that relying solely on local artifacts is insufficient when manipulation techniques vary across regions.
- **ViT Instability:** Similarly, the Federated ViT yields the lowest combined accuracy (92.0%). Without the inductive bias of convolutions, ViTs struggle to converge on forensic features in a decentralized, data-scarce environment.

Our *FedHybrid-ViT* (94.6%) overcomes these limitations by fusing the local artifact sensitivity of the CNN branch with the global semantic reasoning of the ViT branch. This allows the model to detect both pixel-level imperfections and high-level semantic inconsistencies.

5.3 Privacy-Utility Trade-Off

Finally, we address the trade-off between privacy and utility. Typically, federated learning incurs a performance penalty compared to centralized training. However, our results show that *FedHybrid-ViT* achieves 94.6% accuracy, coming within 0.5% of the Centralized Hybrid upper bound (95.1%). This proves that our "Baseline-Federated" training strategy effectively bridges the gap, allowing us to deploy a high-performance deepfake detector that respects user privacy and data sovereignty across international borders.

6 Conclusion

This research introduces a comprehensive privacy-aware deepfake detection framework that employs a hybrid CNN-ViT architecture within a federated learning paradigm to tackle the key challenges of heterogeneous data robustness and statistical variation. By combining the local feature extraction capabilities of Convolutional Neural Networks with the global semantic reasoning power of Vision Transformers, the system effectively captures multi-scale forensic artifacts. Robustly discriminating between authentic and manipulated media from different geographic populations.

Our empirical analysis demonstrates that the proposed **FedHybrid-ViT**model achieves the best performance among the evaluated federated configurations while maintaining performance comparable to centralized hybrid training. Specifically, it achieves an AUC of 0.925 among the evaluated configurations and an accuracy of 94.59%, significantly surpassing conventional federated baselines such as Fed-CNN and Fed-ViT, which experience greater performance degradation under heterogeneous non-IID conditions. More importantly, our "Baseline-Federated" training strategy effectively mitigates the "silo effect," allowing the global model to reach the high performance of specialized regional detectors without compromising data privacy.

Despite these strong results, some limitations persist. The computational complexity of the hybrid architecture remains a challenge for deployment on resource-constrained edge devices. Additionally, although federated learning protects raw data, the system can still be vulnerable to advanced gradient inversion attacks. Future research will focus on integrating lightweight attention mechanisms such as MobileViT, introducing formal privacy guarantees through Differential Privacy (DP) or Secure Multi-Party Computation (SMPC), and incorporating adaptive optimization algorithms to further stabilize training in highly heterogeneous environments.

This work demonstrates the feasibility and impact of applying hybrid deep learning architectures to decentralized forensic tasks. It lays the foundation for developing scalable, privacy-preserving deepfake detection systems capable of enhancing digital trust and security in an increasingly borderless online ecosystem.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare that there are no conflict of interest with respect to the publication of this article.

Ethical Approval All data used in this study were publicly available images; here we have used faceforensics++ and some of our self-made dataset, and no animal subjects were involved.

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